The 2026-05-25 Calama earthquake (M_w 6.9, us6000t04s): moment tensor and finite-fault analysis of an intraslab rupture at 109 km depth

M.-S. Seo

analysis pipeline: MTUQ (point-source moment tensor) + WISP (kinematic finite fault)

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Abstract

We determine the point-source moment tensor (MT) and a kinematic finite-fault model (FFM) for the 2026-05-25 21:52:20 UTC M_w 6.9 earthquake that struck 109 km beneath Calama, northern Chile — an intraslab normal-faulting rupture inside the subducted Nazca plate for which USGS published moment tensors but no finite-fault model. Using 26 teleseismic GSN stations, our final moment tensor (M_w 6.90, strike/dip/rake = 194/16/−83, with conjugate plane 7/74/−92) lies 8.4° (Kagan angle) from the USGS W-phase solution and fits both the 25–60 s body waves (variance reduction 71.9%) and the 50–100 s surface waves (67.2%) better than the USGS mechanism itself evaluated on the same data. The finite-fault model resolves a single compact asperity — on the preferred (shallow) plane a thin slip lens with 0.70 m peak confined to 105–113 km depth, a single ∼15-s moment-rate pulse, subshear rupture — with the conjugate planes indistinguishable in misfit, as expected for a compact deep rupture viewed teleseismically. Depth-phase-sensitive estimates (finite-fault slip centroid 106.5 km; long-period body-wave depth scan 117 km; USGS body-wave MT 111 km; hypocentre 109 km) place the best-constrained centroid depth at ∼105–115 km. Beyond the source parameters, this event provided a controlled experiment in how standard teleseismic source inversion fails at intermediate depth — through depth-phase interference in short-period body-wave windows and through a silent truncation of finite-fault surface-wave Green's functions — and how each failure is diagnosed and fixed. A first-motion polarity anomaly at five near-nodal stations independently discriminates between our mechanism and the USGS solution, favouring ours.

Plain-language summary

A magnitude-6.9 earthquake occurred 109 km below the city of Calama, inside the oceanic (Nazca) tectonic plate that is sinking beneath South America. Earthquakes this deep happen inside the sinking plate, not on the boundary between plates, and are pulled apart by the weight of the plate's deeper portion — our analysis confirms this "slab-pull" stretching almost exactly along the direction the plate is sinking. Using seismometers thousands of kilometres away, we recovered how the fault is oriented, how much it slipped (up to ∼0.9 m over a patch a few tens of km across, in about 15 seconds), and how deep the rupture was. Two technical traps specific to deep earthquakes — both of which initially produced confident but wrong answers — are documented step by step, with the checks that caught them, so that the workflow can be reused by non-specialists on future events.

1 Tectonic setting

Beneath northern Chile the oceanic Nazca plate converges with South America at roughly 7 cm/yr and subducts along the Chile trench, dipping eastward beneath the Atacama forearc and the active volcanic arc (Figure 1). Seismicity here occurs in three distinct habitats: shallow megathrust earthquakes on the plate interface (e.g., the 1995 M_w 8.0 Antofagasta and 2014 M_w 8.2 Iquique events), crustal earthquakes in the overriding plate, and intermediate-depth intraslab earthquakes (70–300 km) inside the subducted plate itself. The 2026 Calama event, at 109 km depth directly below the arc-front longitude, belongs to the third class — the same family as the 2005 M_w 7.8 Tarapacá earthquake ($\sim$95 km deep, normal faulting, $\sim$500 km to the north), which demonstrated that this part of the slab can host large, damaging ruptures far from the plate boundary.

Intermediate-depth earthquakes cannot be explained by ordinary frictional sliding — at these pressures rock should deform by flow, not stick-slip — and are generally attributed to dehydration embrittlement: water released by minerals in the sinking oceanic crust and mantle raises pore pressure and re-enables brittle failure, often re-activating the normal faults created at the outer rise before subduction. Mechanically, the deeper slab hangs off the shallower slab, stretching it down-dip ("slab pull"). Our solution matches this picture quantitatively: the tension (T) axis of the final mechanism plunges 29° toward N98°E — within a few degrees of the local slab down-dip direction — i.e., the slab snapped in down-dip extension. Both nodal planes (a steep east-dipping plane, 7/74, and a gently west-dipping plane, 194/16) are geometrically plausible reactivated structures; as shown in §5, teleseismic data cannot choose between them for a rupture this compact, and we discuss both.

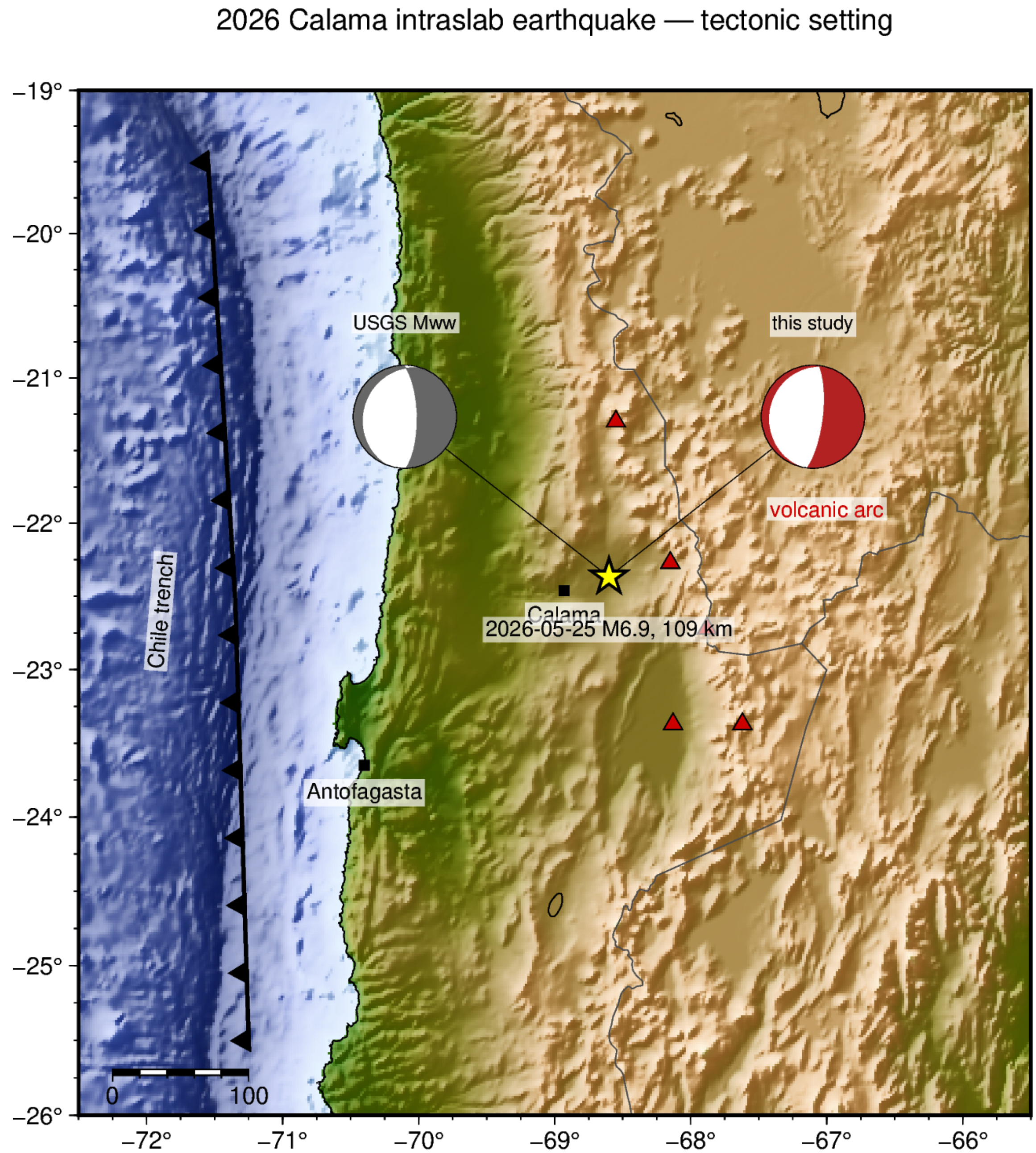

Figure 1: Tectonic setting. The Nazca plate subducts eastward at the Chile trench (toothed line); red triangles mark main-arc volcanoes. The star is the 2026 Calama epicentre (hypocentre 109 km); focal spheres compare the USGS W-phase mechanism (grey) with this study's (red) — both normal-faulting, 8.4° apart.

2 Methods in brief (a primer)

Readers familiar with source inversion can skip to §3.

Moment tensor (MT). A point-source description of an earthquake: the fault orientation (strike/dip/rake), the size (moment M_0, expressed as $\mathrm{M_w}$), and the source type. It is estimated by generating synthetic seismograms for trial mechanisms (using Green's functions — the Earth's response to an impulsive source in a reference velocity model, here ak135) and grid-searching for the mechanism whose synthetics best match the observations. A double couple (DC) is the special case of pure slip on a planar fault; every DC has two mathematically indistinguishable conjugate fault planes, so distinguishing the real plane always needs extra information beyond the radiation pattern. Our searches use MTUQ, which implements the cut-and-paste (CAP) strategy: body-wave and surface-wave portions of each record are windowed, filtered into separate period bands, and allowed small independent time shifts, so that imperfect knowledge of Earth structure does not corrupt the mechanism estimate.

Finite-fault model (FFM). Relaxes the point-source approximation: the fault plane is divided into subfaults, each with its own slip, rake, rupture-arrival time and rise time, inverted from waveforms (here with WISP, the USGS operational code, using teleseismic P, SH, Rayleigh and Love waves). The output is a slip map and a moment-rate function (how fast moment was released versus time).

Depth phases — the key concept at 109 km. Besides the direct P wave, a deep source sends energy up, which reflects off the Earth's surface and follows P down to the same station: the phase pP (and sP for an up-going S leg). The lag pP−P grows with source depth ($\approx$25 s at 109 km) and is the single most depth-sensitive observable in teleseismic data. Crucially, pP reflects with flipped polarity. Whether a method treats these arrivals as signal or lets them contaminate its windows decides whether depth helps or hurts — the central methodological theme of this study.

Fit and comparison metrics. Variance reduction (VR) is the percentage of waveform energy explained by the synthetics (100% = perfect; negative = worse than predicting zero motion). The Kagan angle is the single 3-D rotation between two mechanisms (0° = identical; routine agency solutions for the same event typically differ by 10–20°). WISP reports a normalized waveform misfit (lower is better); misfits are comparable only between runs inverting the same data with the same wave types.

3 Data and processing

MTUQ: 26 three-component GSN broadband stations at 25–75° epicentral distance, azimuthally distributed. Instrument response was removed to ground displacement with no water-level stabilization (a water level biases long-period amplitudes — a lesson from the companion Venezuela study), data at 2 Hz. WISP: 180 teleseismic channels fetched with an obspy-based fallback downloader (`wisp_fetch_fallback.py`; the packaged IRIS discovery endpoint is no longer served), pole-zero response files renamed and headers fixed with true BH1/BH2 azimuths. Deep-event P is impulsive: $\sim$45–49 P channels pass WISP's signal-to-noise gate — in sharp contrast to a same-magnitude shallow event (our Sanriku study), where all P channels failed it. Three surface-wave channels that no model variant could fit (EDA Rayleigh and Love, PTCN Love — isolated island paths) were zero-weighted in the final inversion and are shown grey in the fit figures. Figure 2 shows both station sets. All inversion parameters are listed in Appendix A.

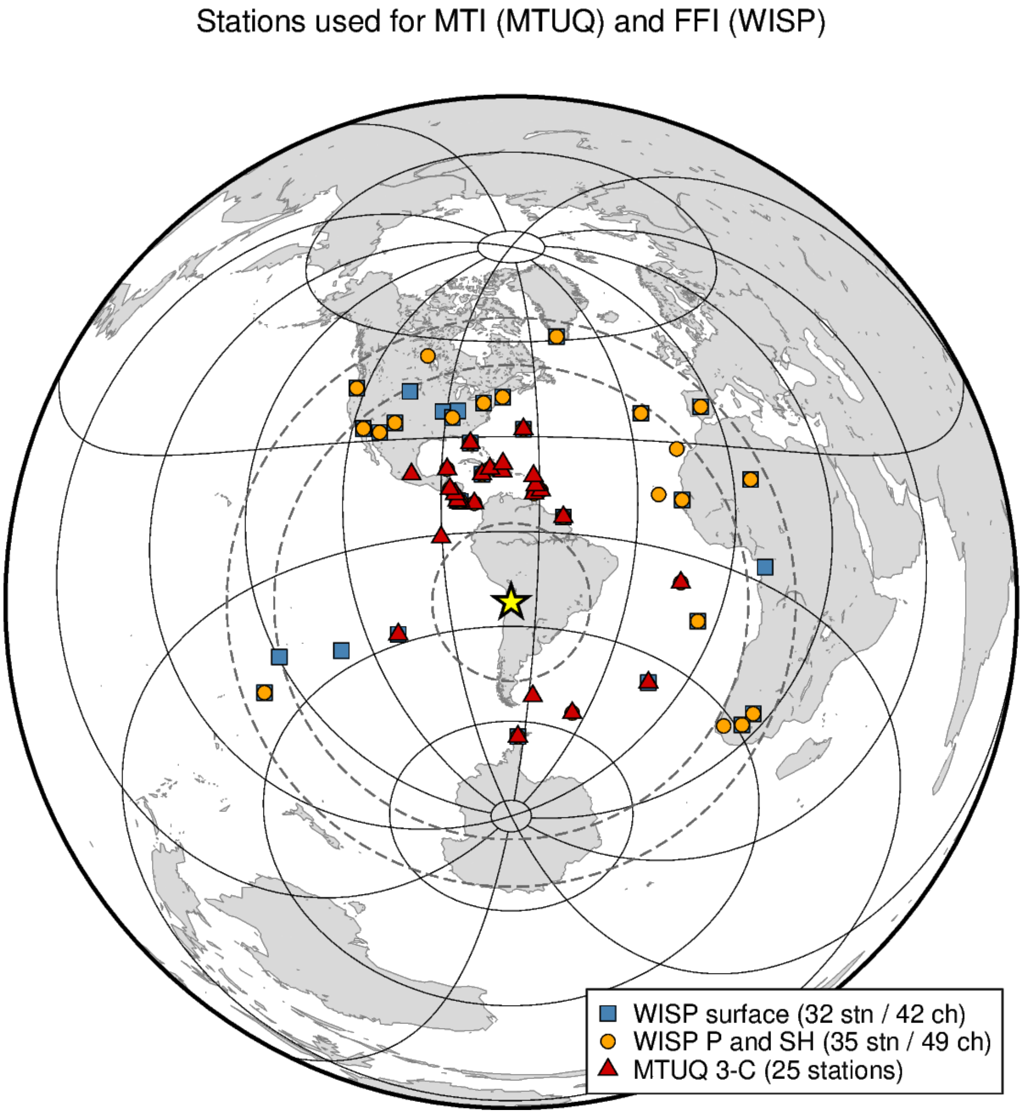

Figure 2: Stations used. Red triangles: the 25 GSN three-component stations of the MTUQ inversions (dashed circles: the 25–75° selection window). Orange circles / blue squares: WISP body-wave (35 stations, 49 P+SH channels) and surface-wave (32 stations, 42 channels; 39 active in the final model) sets. Azimuthal-equidistant projection centred on the epicentre (star).

4 Moment tensor: failure, diagnosis, cure

Table 1 traces the full sequence — we report it in order because the failure modes are as reusable as the final numbers.

run	band	M_w	strike/dip/rake	Kagan vs Mww	note
full MT @109 km	body 10–33 s + SW	6.80	54/18/−45 (w=0.905)	—	strongly non-DC: red flag
DC @109 km	body 10–33 s + SW	6.75	4/75/+83	85.5°	reverse — the body flip
DC + depth (85–125 km)	body 10–33 s + SW	6.75	7/75/+79 @85 km	85.3°	same flip; edge of range
DC + depth (55–115 km)	SW only	6.90	208/19/−64 @115 km	3.3°	monotonic: no depth resolution
DC @85 km	SW only	6.90	14/75/−83	17.7°	≈ USGS Mww
DC @107 km	body 25–60 s only	6.90	7/82/−72	26.7°	body waves recovered
DC + depth (85–125 km)	body 25–60 s only	6.90	7/80/−86 @117 km	14.9°	interior depth minimum
DC + depth (85–125 km)	body 25–60 s + SW	6.90	209/20/−65 @125 km	∼3°	edge; SW dilutes depth signal
DC @107 km (final)	body 25–60 s + SW	6.90	194/16/−83	8.4°	all data; 107 km adopted (= WISP ce
USGS Mww	W-phase	6.9	359/71/−97 @85.5 km	0	reference

Table 1: Moment-tensor runs, in chronological order. SW = surface waves (50–100 s). Kagan angles computed with the canonical `pyrocko` implementation. w is MTUQ's non-DC lune coordinate.

4.1 The failure

With the standard teleseismic settings (body waves at 10–33 s in a 40-s window starting at the P arrival), the DC search converged — reproducibly, with a clean-looking misfit surface — on a reverse-faulting mechanism ∼85° from every published solution. Data hypotheses (time-shift clamping; a non-band-limited resampler) were tested and excluded: freshly re-prepared, Lanczos-resampled data still flipped. The decisive measurement was evaluating the USGS mechanism on our own body windows: VR = −24%. When the accepted answer fits worse than a flat line, the protocol — not the Earth, not the data — is broken. (The unconstrained full-MT search absorbed the same inconsistency into a nearly pure non-double-couple source, $w = 0.905$: at depth, a strongly non-DC result should be read as a symptom, not a discovery.)

4.2 The diagnosis and the cure

At 109 km the 40-s body window contains P, pP and sP together, and the depth phases carry 3–5× the direct-P energy with flipped polarity (visible directly in Figure 11). Their spacing inside the window is set by source depth, not by the window's single allowed time shift; and because the rupture spans ∼33 km of depth, the observed "pP" is really a ∼10-s smear that no single point depth reproduces. A point source with a simple wavelet is valid only at periods long compared to both the ∼15-s source duration and that smear ($T \gtrsim 45$ s), while resolving pP as a distinct pulse requires $T \lesssim 25$ s: for an M_w 6.9 at 109 km there is no band where a short-period point-source treatment both fits the body waves and resolves depth. The practical cure is to move the body band to 25–60 s in a 120-s window that holds the whole P+pP+sP group as one unit (±10 s shifts, Green's functions at the 107-km centroid). This single change flips body-wave VR from −80.5% to +71.9% (our mechanism) and −24.0% to +67.7% (USGS), demotes the spurious reverse mechanism to 18.5%, and lets a body-only search recover the correct normal-faulting family. Unexpectedly, the long-period body waves also retain interference-shape depth sensitivity (the composite P+pP+sP waveform changes shape with depth even though pP is not separately resolved, as in W-phase methods): a body-only depth scan exhibits a genuine interior misfit minimum at 117 km (Figure 3, right).

4.3 The final solution

The joint search (25–60 s body + 50–100 s surface waves, 107 km) yields M_w 6.90, 194/16/−83 (conjugate 7/74/−92), 8.4° from the USGS Mww — well within routine inter-agency scatter — with VR of 71.9% (body) and 67.2% (surface waves), each higher than the USGS mechanism evaluated on identical data and Green's functions (67.7%/66.7%). The first-motion evidence of §5.2 independently favours this small rotation away from the USGS solution.

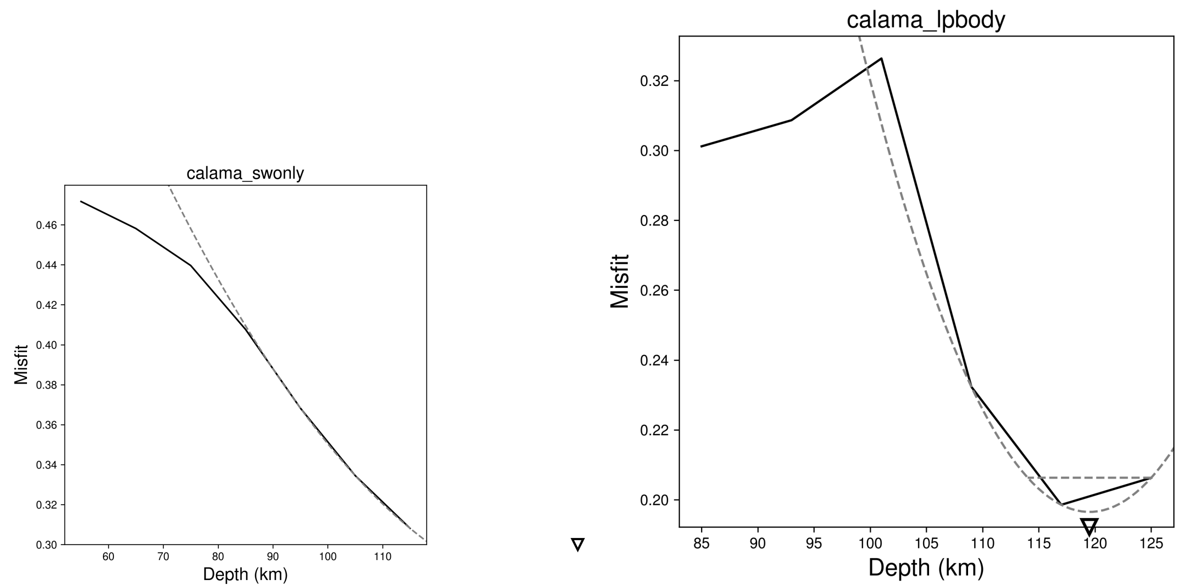

Figure 3: DC misfit vs depth. Left: surface waves only (55–115 km) — monotonic; 50–100 s surface waves carry no centroid-depth resolution for a source this deep. Right: body waves only, 25–60 s band (85–125 km) — a genuine interior minimum at 117 km.

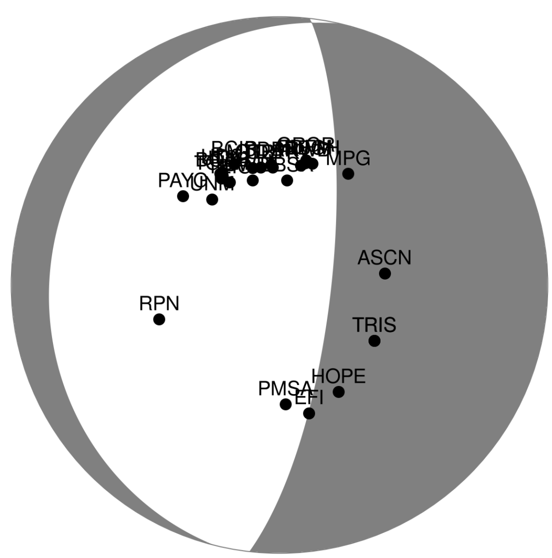

Figure 4: Final DC (194/16/−83; body 25–60 s + surface waves, 107 km) and the 26 stations.

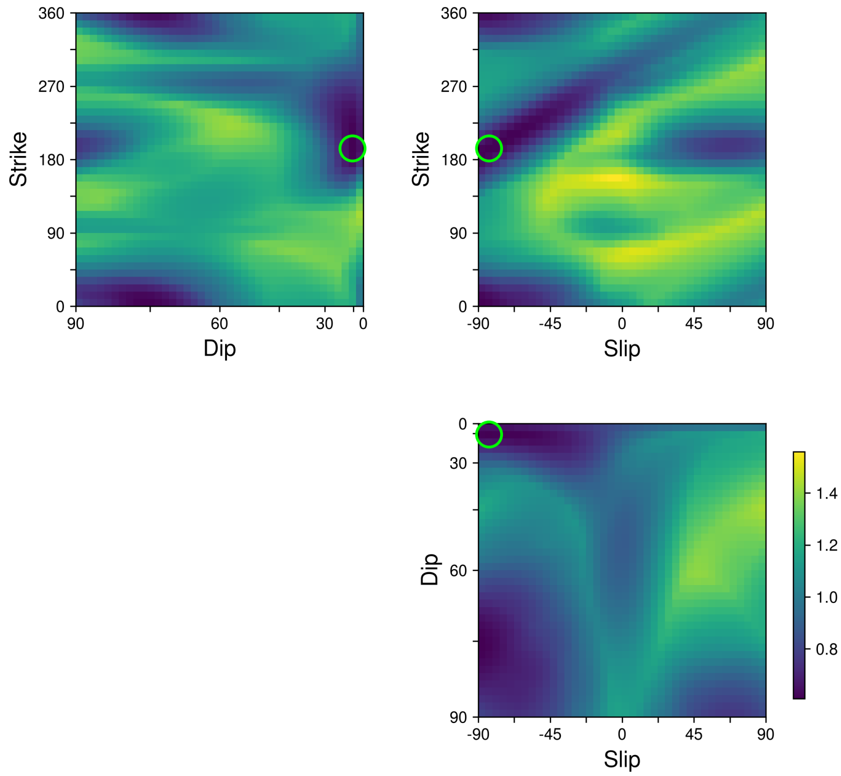

Figure 5: DC misfit surfaces (strike–dip–rake) for the final joint search.

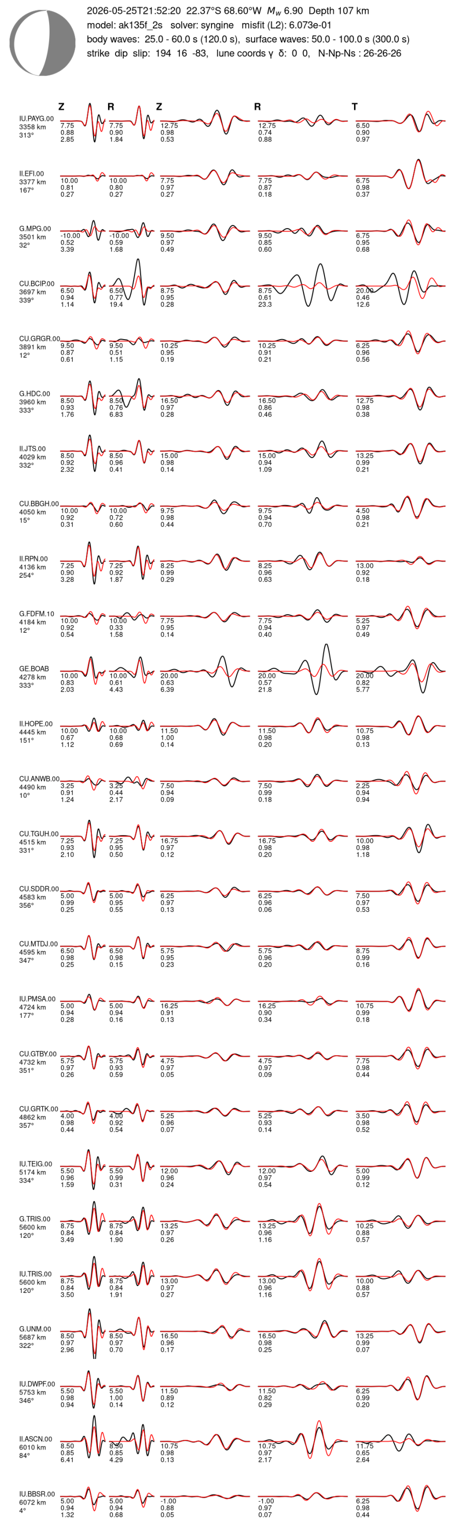

Figure 6: Waveform fits at the final DC (black = observed, red = synthetic): 25–60 s body windows (left columns) and 50–100 s surface-wave windows.

5 Finite-fault model

WISP was run in its strict CLI pipeline (automatic model construction, both nodal planes, teleseismic body + surface waves). Two findings shaped the analysis.

A silent Green's-function truncation manufactured a fake plane discrimination. The steep plane's automatically built fault (15 down-dip rows) reached 133.6 km — below the ~126-km floor of WISP's surface-wave Green's-function bank — and WISP dropped all surface waves for that plane only, noting it in a single log line. The resulting misfits (0.114 for the steep plane vs 0.166 for the shallow plane) compared a body-only number against a body+surface number and made the steep plane appear strongly preferred. After trimming the fault to 12 rows and refining both planes identically (all quality control and cross-correlation alignment applied equally, all wave types inverted), the planes are indistinguishable:

input mechanism	plane	misfit	peak slip	slip depth extent	slip centroid
USGS Mww	shallow (201/20)	0.1470	0.75 m	104–115 km	108.8 km
USGS Mww	steep (359/71)	0.1473	0.78 m	88–121 km	106.2 km
this study	shallow (194/16)	0.1461 (0.1402†)	0.70 m	105–113 km	109.3 km
this study	steep (7/74)	0.1481	0.89 m	94–121 km	106.8 km

Table 2: The fair model comparison: two input mechanisms × two nodal planes, all with identical fault trimming, quality control, channel set (49 P + 42 surface) and alignment; all M_w 6.93. Misfit differences are at the 10^{-3} level — individually not decisive — but the best-fitting model is the shallow plane with this study's mechanism, concordant with the regional precedent (§5). †Final value after zero-weighting the three unfittable island-path surface channels (EDA, PTCN).

Both planes carry the same double couple (≤2° from the USGS Mww). Which is the real fault? The waveforms cannot say, but three physical arguments can be weighed. (i) The closest regional analog, the 2005 Tarapacá M_w 7.8 (same slab, ~95 km), was shown by aftershock geometry and rupture extent to have ruptured its near-horizontal plane — the NP1-type geometry. (ii) NP1's slip forms a thin subhorizontal lens (104–115 km, only ~11 km thick) that fits comfortably inside the slab's brittle core, whereas NP2's slip spans 33 km vertically — close to the full seismogenic thickness of the slab at this depth. (iii) The rotated outer-rise-fault argument permits both: outer-rise normal faults form as conjugate pairs, and subduction rotation carries the trenchward-dipping set to NP2-like steep dips and the seaward-dipping set to NP1-like shallow dips. On balance, regional precedent favours the shallow plane, and — weakly but concordantly — so does the fit: with this study's mechanism the shallow plane gives the best misfit of all four models (Table 2). We adopt the shallow plane (194/16) as the preferred rupture plane, show both, and note that a definitive answer needs locally recorded aftershocks.

Slip model. A compact rupture centred essentially at the hypocentre. The preferred model (shallow plane 194/16, this study's mechanism): a thin subhorizontal slip lens — peak slip 0.70 m, slip (above 15% of peak) confined to 105–113 km depth, ~30 km along strike, slip-weighted centroid 109.3 km, one moment-rate pulse of ~15 s effective duration, subshear rupture, rise times 2–7 s, M_w 6.93 (moment 3.17×10^{19} N m). The steep-plane alternative (7/74) distributes the same moment over 94–121 km of depth with 0.89 m peak; all four mechanism/plane variants agree on the moment, duration and centroid depth to within a few km (Table 2).

5.1 Source dimensions and stress drop

From the slip models directly (rigidity $\mu = 67.5$ GPa from the inversion's own shear model at source depth; effective dimensions from a slip threshold; Eshelby circular-crack $\Delta\sigma = \frac{7}{16} M_0^{\mathrm{eff}}/R^3$, $R = \sqrt{A_{\mathrm{eff}}/\pi}$):

model (threshold)	A_{eff}	L	W	$\bar{D}$	M_0^{eff}/M_0	$\Delta\sigma$
shallow, final (15%)	1093 km^2	39 km	35 km	0.34 m	0.90	1.7 MPa
shallow (30%)	744 km^2	34 km	28 km	0.43 m	0.77	2.6 MPa
steep (15%)	850 km^2	34 km	32 km	0.44 m	0.90	2.5 MPa
steep (30%)	577 km^2	30 km	25 km	0.55 m	0.77	3.8 MPa

Table 3: Effective source dimensions and static stress drop. A width-limited estimate ($\Delta\sigma \approx 2.5\,\mu\bar{D}/W$) agrees with the Eshelby values to within 0.2 MPa in every row.

Reading and caveats. (i) Smoothed teleseismic finite-fault models systematically overestimate rupture area, so these $\Delta\sigma \approx 1.7$–2.6 MPa (preferred plane; threshold sensitivity shown) are best read as lower bounds of order a few MPa. An independent, smoothing-free check supports the moderate size: the $\sim$10–15 s moment-rate duration with $V_r \approx 2.5$ km/s implies $R \approx 13$–19 km, matching the model's effective radius — so the source really is $\sim$30–40 km across, and the stress drop is genuinely of order a few MPa, not tens. (ii) Against global scaling: at $\mathrm{M_w}$ 6.9 the Wells & Coppersmith (1994) normal-fault relation (crustal) predicts $\sim$650 km^2; our effective areas (740–1100 km^2) sit modestly above, i.e., this event is ordinary to slightly low in stress drop relative to crustal norms and clearly not in the elevated tens-of-MPa class reported by spectral studies for many intermediate-depth events — consistent with its unremarkable rise times (2–7 s) and subshear rupture velocity. (iii) Aspect ratio: $L/W \approx 1.1$, an equant rupture. Crustal normal faults at this magnitude are width-limited by the seismogenic layer ($W \lesssim 15$–20 km, forcing $L/W \gtrsim 2$); an intraslab rupture at 107 km has no such free-surface constraint, and the near-circular shape is the expected geometry — a small but tidy consistency argument for the intraslab setting.

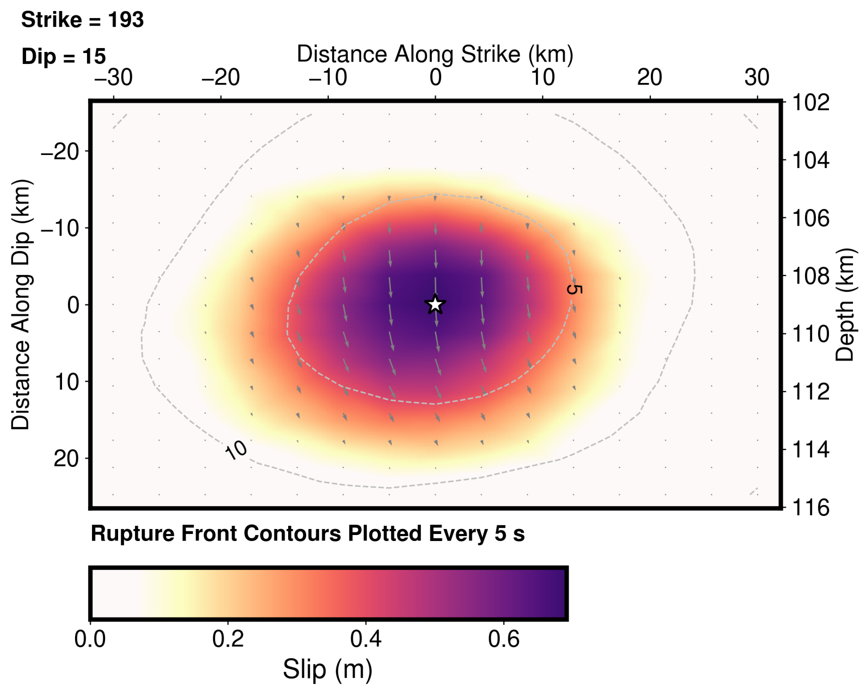

Figure 7: Preferred slip model (shallow plane 194/16, this study's mechanism; WISP native style; white contours = rupture front every 5 s).

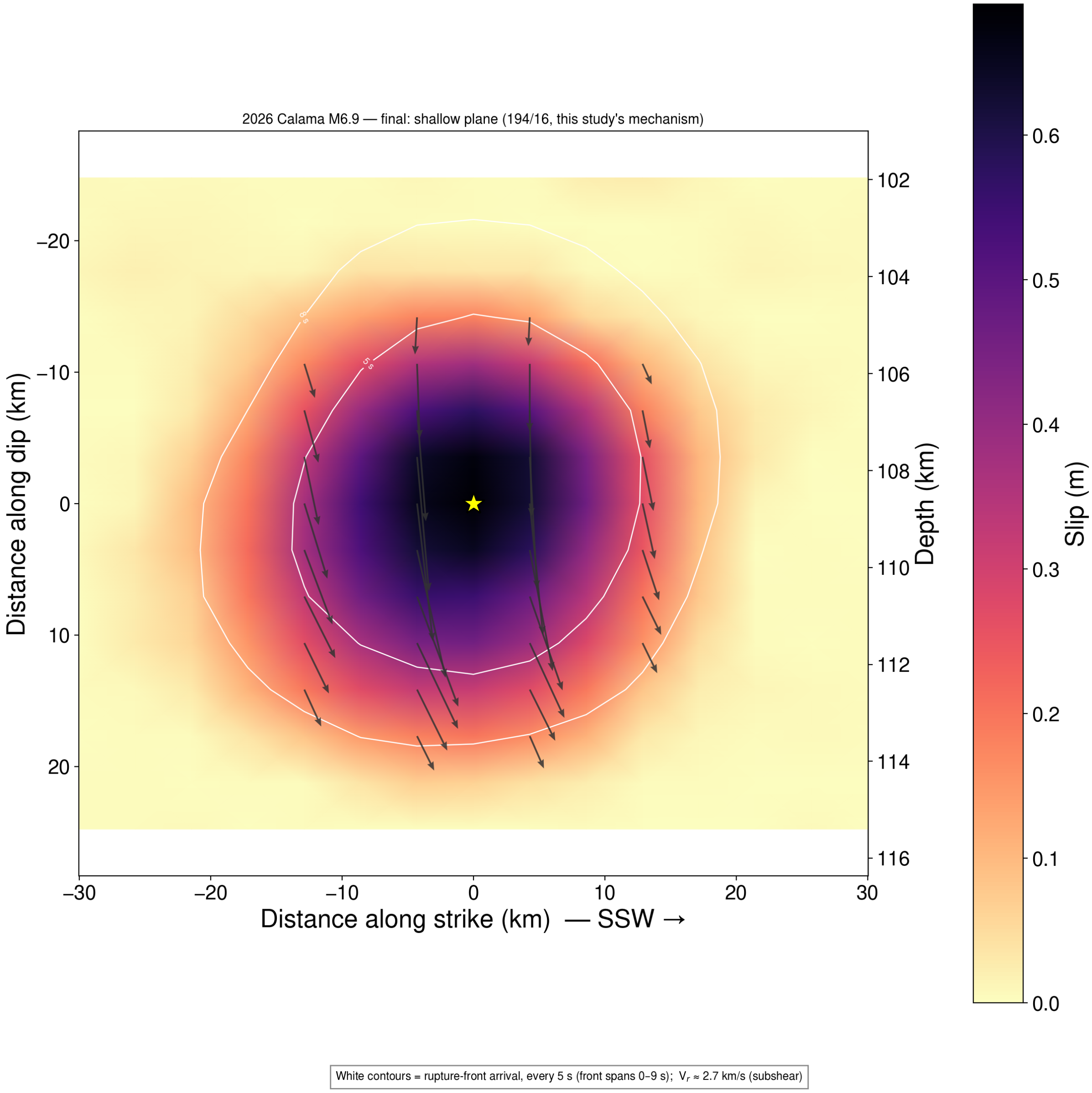

Figure 8: True-scale (1:1) slip, slip vectors and rupture isochrones on the preferred shallow plane; the right axis converts along-dip position to depth. Note the thin (∼8 km) subhorizontal slip lens — the Tarapacá-style geometry.

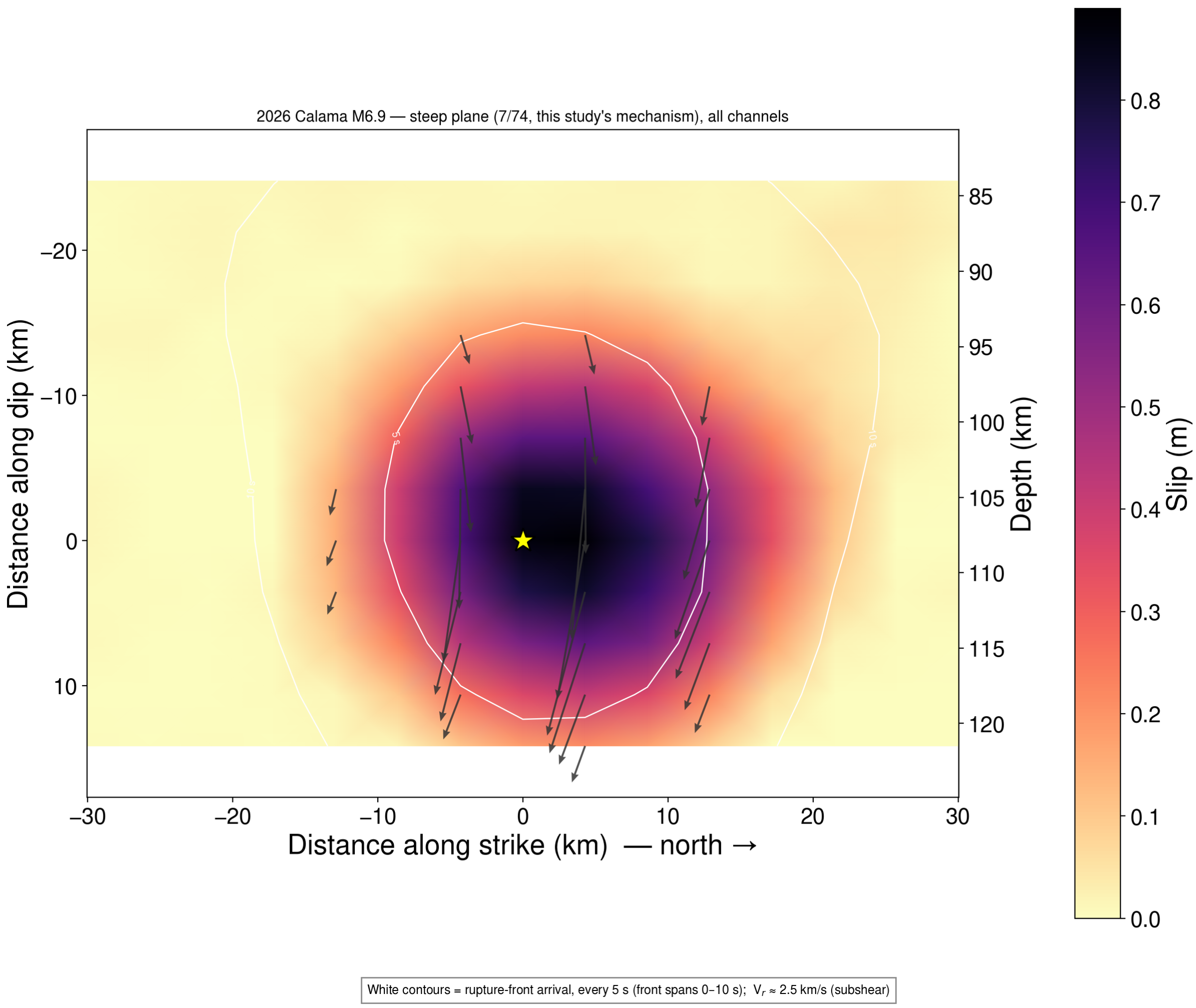

Figure 9: The conjugate model (steep plane 7/74), shown for completeness: nearly equal misfit, same moment and mechanism, mirrored geometry.

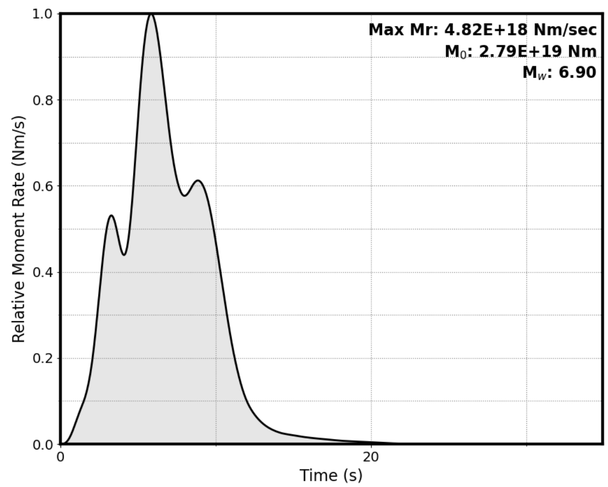

Figure 10: Moment-rate function: a single pulse peaking at ∼7 s, effective duration ∼15 s.

5.2 Waveform fits and the first-motion finding

The P-wave figure (Figure 11) is the deep-event story in one panel: the direct P (∼5 s) is small, while the depth-phase group pP/sP (30–50 s) carries several times more energy and is well modelled — the same energy that, unmodelled, flips a short-period point-source search (§4).

Close inspection of these fits exposed a subtler, more interesting defect: at five high-quality stations with clean positive first motions (MACI, TAM, LBTB, BOSA, SUR — African/Atlantic azimuths, 47–120°), the initial pulse is not modelled at all, while the later depth phases fit well; every negative-first-motion station fits throughout. Radiation-pattern analysis explains it: for the USGS Mww mechanism these five stations lie almost exactly on the P nodal surface (predicted amplitudes +0.02 to +0.06, versus −0.6 to −0.85 at the well-fit stations) — WISP, whose fault-plane orientation is fixed by its input mechanism, was faithfully reproducing an input that predicts near-zero direct P there. The observed robust positive onsets require the mechanism to be rotated a few degrees — and our final MTUQ mechanism does exactly that, raising the predicted amplitudes at those stations by a factor of 3–6 (+0.06 to +0.22) while leaving the negative-polarity stations unchanged. First motions at near-nodal stations thus independently discriminate between two mechanisms only 8.4° apart, in favour of ours. The direct test — a finite-fault re-inversion using our moment tensor as the input mechanism (planes 194/16 and 7/74, identical trimming, quality control and alignment) — passes: the previously missing initial pulses appear at all five stations with correct polarity and timing, at comparable-to-better overall misfit (best model 0.1461; Table 2). Amplitudes remain somewhat under-predicted, consistent with residual mechanism rotation and/or unmodelled slab-path effects on the near-nodal direct P.

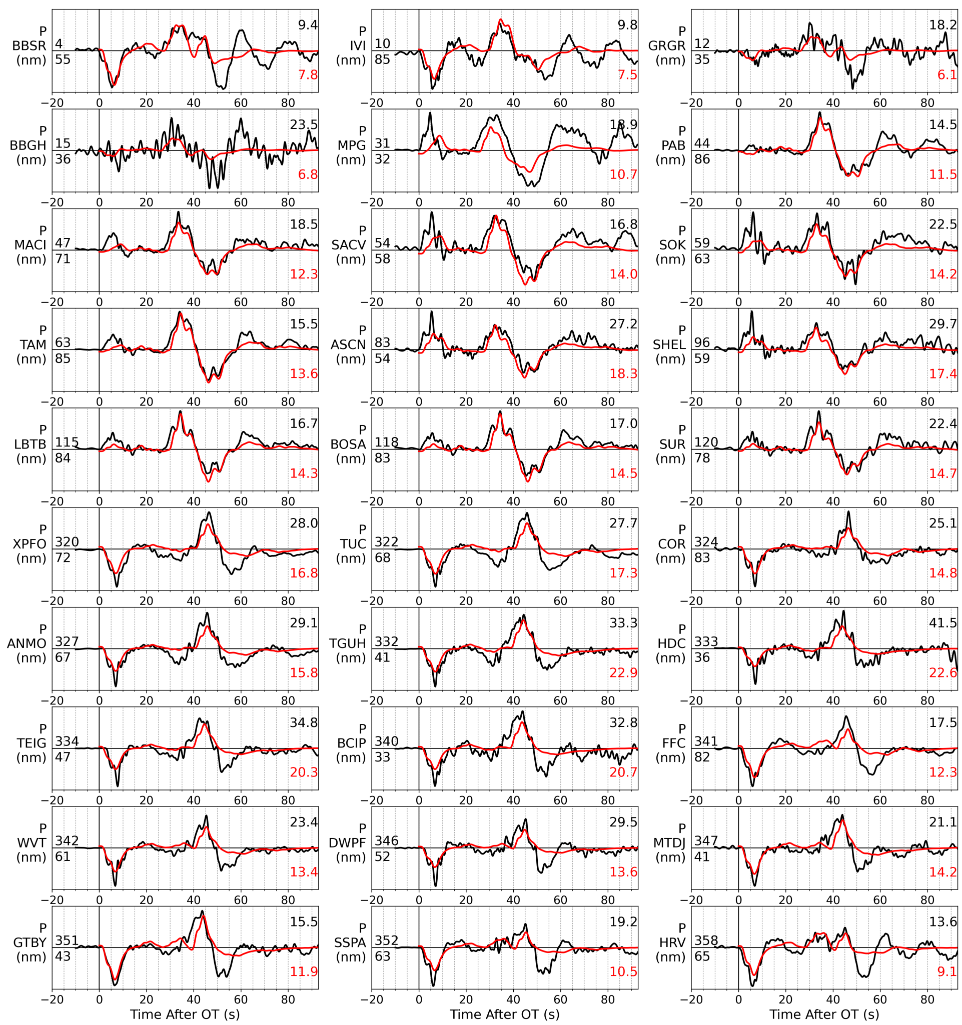

Figure 11: P-wave fits (preferred model). Note (i) the dominant, well-modelled depth-phase group at 30–50 s, and (ii) the clean positive onsets at MACI, TAM, LBTB, BOSA and SUR that the USGS-mechanism-based model cannot produce (§5.2).

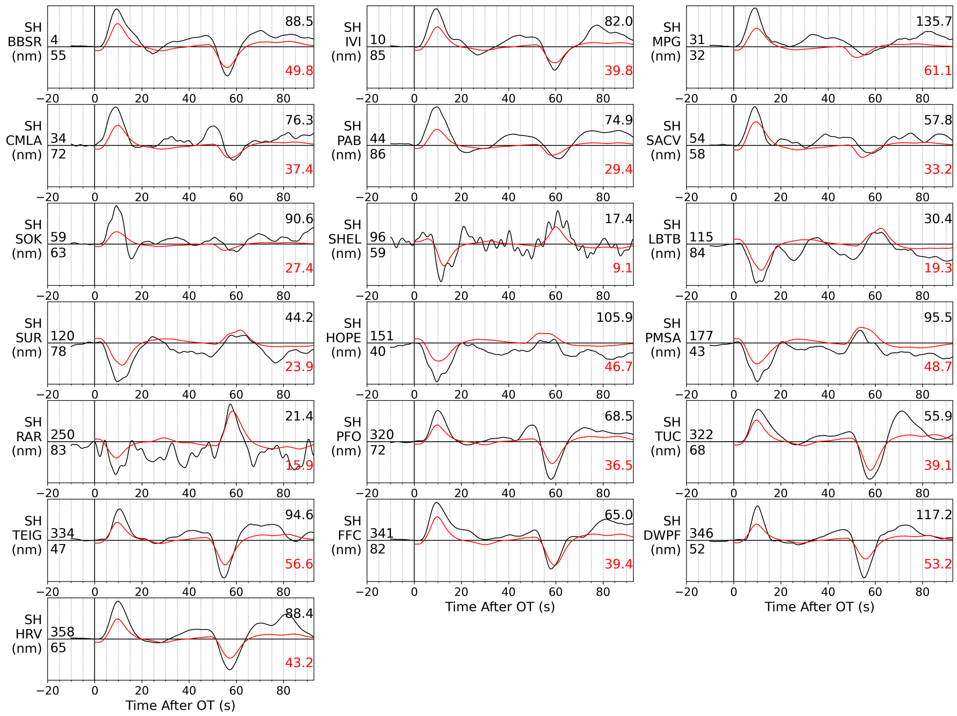

Figure 12: SH-wave fits (preferred model).

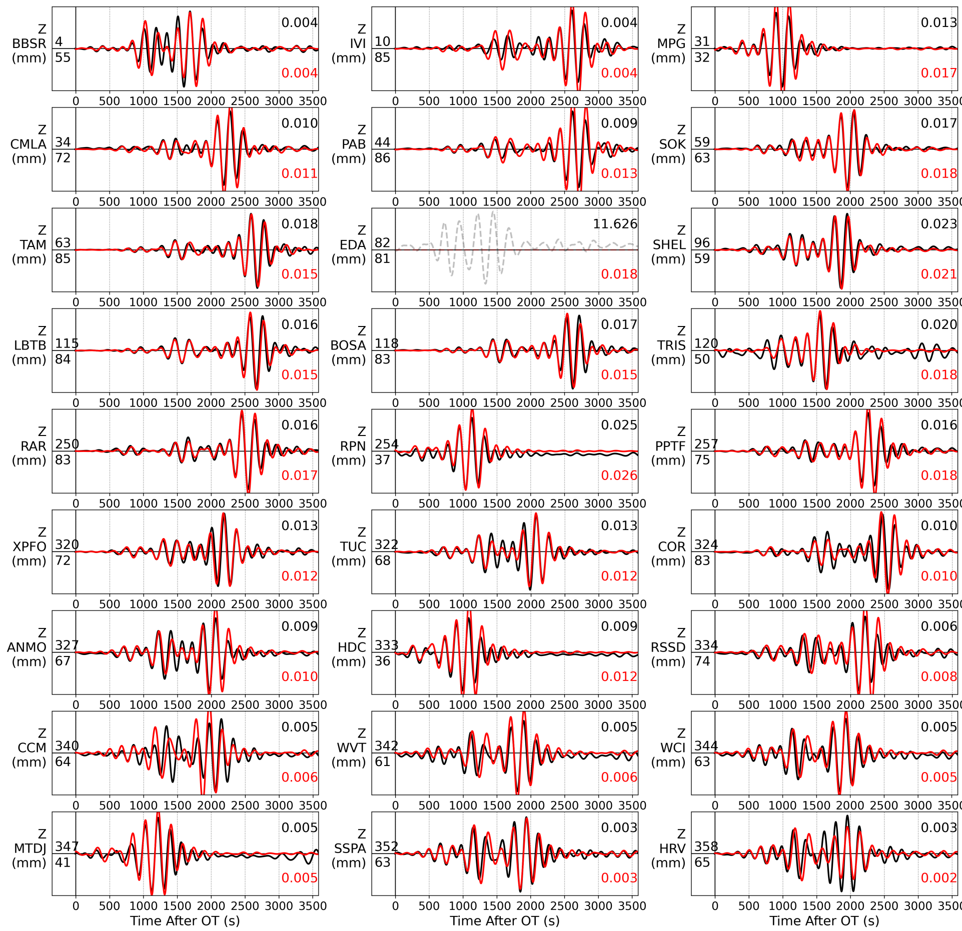

Figure 13: Rayleigh-wave fits (preferred model).

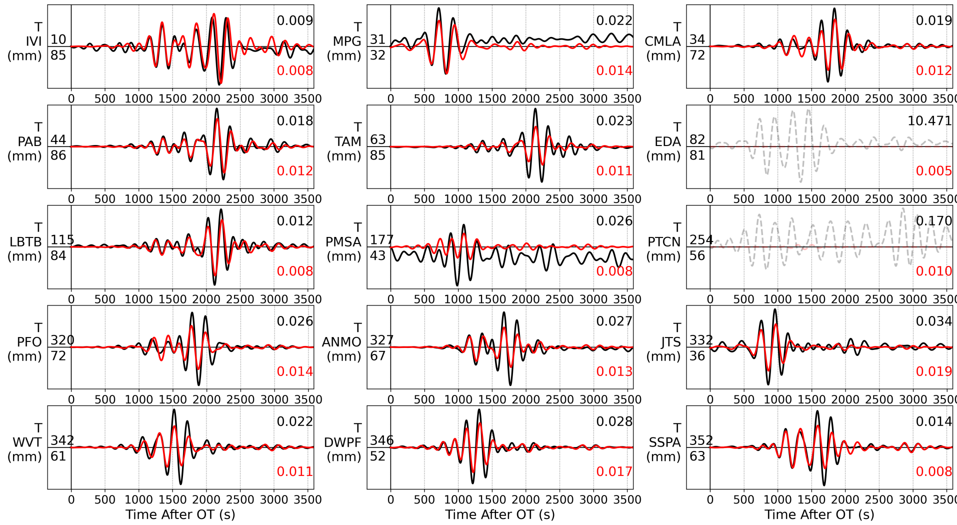

Figure 14: Love-wave fits (preferred model).

6 How deep was it, really?

No single method resolves the centroid depth here, but four depth-phase-sensitive estimates — obtained by different physics and different codes — cluster tightly:

estimate	depth	depth information carrier
WISP slip-weighted centroids	106.2–108.8 km	pP/sP timing per subfault[a]
arrival-time hypocentre (USGS)	109 km	first-arrival times
USGS body-wave MT (Mwb)	111 km	depth phases
MTUQ body-only depth scan (25–60 s)	117 km	P+pP+sP interference shape
USGS Mww / Mwc (long-period)	90.5 / 85.5 km	weak at this depth
our SW-only depth scan	unresolved	monotonic misfit — no minimum

Table 4: Centroid-depth estimates. [a]The WISP fault grid spans 84–127 km, so the inversion was free to place slip near 85–90 km and did not; note the grid is anchored at the 109-km hypocentre (hypocentre-anchored but depth-phase-refined).

We conclude the centroid lies at ∼105–115 km. Three reading notes. First, no MTUQ misfit curve has its minimum at our evaluation depth of 107 km: that number is the WISP slip centroid, adopted (and labelled as such) because the finite-source treatment of the depth phases is the most physically complete estimator available — the choice has no effect on mechanism or magnitude, which are stable from 55 to 125 km. Second, the MTUQ body-only minimum at 117 km is expected to sit deep: a single point source matching the composite depth-phase smear of a rupture whose slip spans 94–121 km lands near the deep edge of the slip zone, not at its centroid. Third, a joint-band (body+surface) depth scan is worse than the body-only scan for depth — the depth-blind surface-wave term dilutes the body signal and drags the minimum to the deep edge of the searched range — so depth should be read from the body-only curve. The long-period estimates (85–90 km) are not independent evidence of a shallower source: that band carries no depth resolution for this event (Figure 3, left). Finally, the adopted evaluation depth was verified to be inconsequential by direct test: re-running the identical joint search at 117 km yields 212/16/−65 — 5.0° from the 107-km solution and the same 8.4° from the USGS Mww, at the same M_w 6.90.

7 Scientific significance and outlook

Is this event special? Northern Chile hosts the most active intermediate-depth seismic zone in South America — including a documented double seismic zone — so moderate intraslab earthquakes are routine here. Events of $M_w \geq 6.5$ at 100–110 km directly beneath the arc are, however, decadal-scale occurrences, and each well-recorded one is a rare direct probe of the stress state and failure mechanics inside the slab, where no geodetic or geologic observable reaches. This event adds three specific data points. (i) Stress state: the T-axis is aligned with the slab down-dip direction to within a few degrees — clean slab-pull extension at 109 km beneath the arc, a boundary condition for models of slab coupling and unbending in this segment. (ii) Failure mechanics — where plane discrimination carries real physics: the two candidate planes imply different answers to how intermediate-depth faults slip. A steep plane would indicate reactivation of the trenchward-dipping outer-rise fabric (the "textbook" dehydration-embrittlement picture); the shallow plane — weakly preferred here by fit and strongly by the Tarapacá 2005 precedent — implies slip either on the rotated seaward-dipping conjugate set or on newly-formed subhorizontal shear zones (as proposed for Tarapacá), i.e., that the subhorizontal orientation is systematically the weak one in this slab. A growing regional pattern of subhorizontal intraslab ruptures would be mechanically significant and testable event by event. (iii) Hazard: intraslab events are efficient high-frequency radiators (Tarapacá 2005 caused damage despite its 95-km depth), and the finite-fault dimensions and stress drop (§5.1) anchor regional shaking estimates for the mining region around Calama. Note that although the two planes differ strongly in slip depth extent (8 vs 27 km), their along-dip widths and areas are comparable (the shallow plane's 8 km of depth spans ∼30 km measured on the 16°-dipping plane), so the stress drop is nearly plane-independent.

Improving the models with regional data. The event sits under one of the best-instrumented forearcs on Earth: the open IPOC network (CX, ∼20 broadband + strong-motion stations at 21–24°S) plus the Chilean national network (C1) surround the epicentre at 50–300 km. Three concrete upgrades are feasible with open data alone: a regional-waveform moment tensor (MTUQ with frequency–wavenumber Green's functions from a local 1-D model) as an independent mechanism/depth check; finite-fault re-inversion adding regional strong motion (WISP supports it operationally), which is the observable most capable of discriminating the nodal planes — the two planes predict different near-field ground-motion patterns even when teleseismic fits are identical; and a locally recorded aftershock alignment, the classical plane discriminator.

Geodesy is not the tool for this event. Static surface displacement scales as $\sim M_0/(4\pi\mu d^2)$: for $M_0 = 3.2 \times 10^{19}$ N m at $d \approx 107$ km this is ∼1–3 mm spread over a ∼100-km wavelength — below InSAR sensitivity for long-wavelength signals (orbital and tropospheric ramps dominate) and at the margin of continuous GNSS. Geodetic constraints on intraslab sources become practical roughly an order of magnitude larger in moment: the 2005 Tarapacá M_w 7.8 (∼30× this moment) was measurably recorded by regional GPS. For $M_w \lesssim 7$ intermediate-depth events, seismology — teleseismic plus regional — carries all the information.

8 Conclusions and reusable lessons

1. Source parameters. M_w 6.90 normal-faulting intraslab rupture; mechanism 194/16/−83 (conjugate 7/74/−92), 8.4° from USGS Mww with better fits on both wave types; T-axis down-dip (slab-pull extension); centroid ∼105–115 km; compact single asperity (preferred shallow plane: 0.70 m peak, 105–113 km), ∼15 s, subshear; M_w 6.93 from the finite fault. Regional precedent (Tarapacá 2005) and the best fit both point — weakly — to the shallow plane.

2. Conjugate planes are not discriminated by teleseismic data for a compact deep rupture — an apparent discrimination in the automatic runs was a Green's-function artefact. Plane claims for such events need regional data or aftershocks.

3. Deep-event body waves are usable, at the right period. Short-period point-source windows that mix P/pP/sP fail confidently; at 25–60 s the same machinery fits well and even retains depth sensitivity. Check any protocol by evaluating the accepted reference mechanism on your own data — a negative VR for the "right" answer means the protocol, not the Earth, is wrong.

4. Silent failures are the dangerous ones. Both traps here (depth-phase window contamination; the surface-wave Green's-function depth cap) produced clean-looking inversions of the wrong problem. The two habits that caught them: compare only like with like (same wave types, same channels), and read the pipeline log, not just the result.

5. First motions at near-nodal stations are a sharp, free diagnostic — here they discriminated mechanisms 8.4° apart and identified which of two published-quality solutions the data prefer.

A Reproducibility: runs and parameters

All analysis lives in `~/works/17.Venezuela_2026` (scripts in `scripts/`, results in `results/`, WISP working directory `event_2026_calama/`); the companion notebook is `notebooks/24_calama_m69_deep_mt_ffi.ip` and the step-by-step generic recipe is `notebooks/23_mtuq_wisp_cookbook.ipynb`.

MTUQ (final run). `mpirun -n 20 python scripts/15_run_mtuq_dc.py --event calama_lpbody --depth 107 --mag 6.9 --bwband 25,60,120,10`. Grid: `DoubleCoupleGridRegular`, 40 points per axis × 5 magnitudes (320,000 sources). Body waves: bandpass 25–60 s, 120-s window from the ak135 P time, L2 misfit, Z+R components sharing one time shift per station, ±10 s. Surface waves: 50–100 s, 300-s window, Z+R and T groups, ±20 s. Green's functions: syngine ak135; source wavelet: trapezoid scaled to M_w 6.9. Data tagged as displacement; response removed with `water_level=None`; 26 stations, 25–75°. Depth scans: 55–115 km (SW-only) and 85–125 km (body-only, `--onlybody`), `16_run_mtuq_dc_depth.py`. VR: un-normalized L2 through MTUQ's misfit engine at fixed source (`17_mtuq_vr_calama.py`); Kagan angles: `pyrocko.moment_tensor.kagan_angle`. Earlier runs (Table 1): default band `bands(6.9)` = body 10–33 s / 40-s window / ±4 s; full-MT grid `FullMomentTensorGridSemireg`

WISP. Data: `wisp_fetch_fallback.py <time> <lat> <lon>` (obspy; 180 channels); headers fixed with `wisp_fix_headers.py`; auto model: `ffm model run . auto_model -g <CMT> -d . -t body -t surf` for the USGS CMT and again for our MTUQ mechanism (`mtuq_cmt_CMT`). Refinement (`rerun_calama_refine.py <plane> keepresp`): copy the auto plane, trim down-dip rows until the fault bottom clears the 126.5-km surface-wave Green's-function floor (15→12 rows for the steep plane), cross-correlation alignment (`shift_match2`) for body and surface waves, re-invert; all 49 P + 42 surface channels active (an earlier generic-response heuristic that zeroed 4 stations was found too aggressive and disabled — their gains are consistent with neighbours and their fits are good). Fault: 15×12 subfaults of 3.5×3.5 km; simulated annealing as shipped; velocity model: WISP default (ak135-based) at source, PREM-based Green's-function banks.

Known pitfalls encoded in the scripts (each cost real time here): syngine rejects 1-Hz data; MTUQ's resampler has an npts-dependent off-by-one (trim to 12800 samples at 4 Hz); resample only with band-limited interpolation (`lanczos`); MTUQ misfit arrays are (sources, origins) — origins last — when reshaping; MPI `grid_search` returns `None` on non-root ranks; WISP silently drops surface waves when the fault bottom exceeds ∼126 km (grep the pipeline log for "Maximum depth"); `save_waveforms` after `shift_match` can drop the P block from `tele_waves.json`.